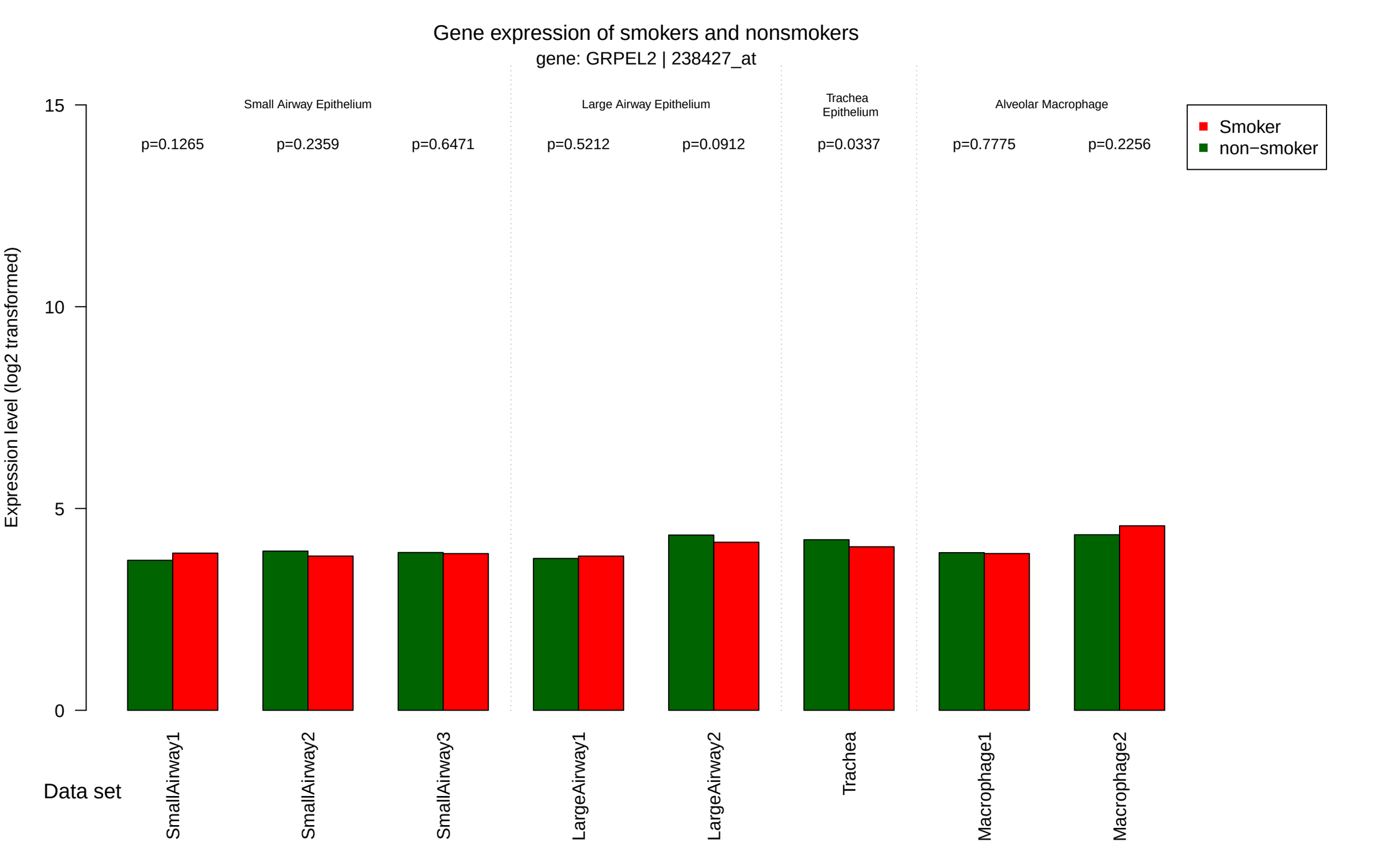
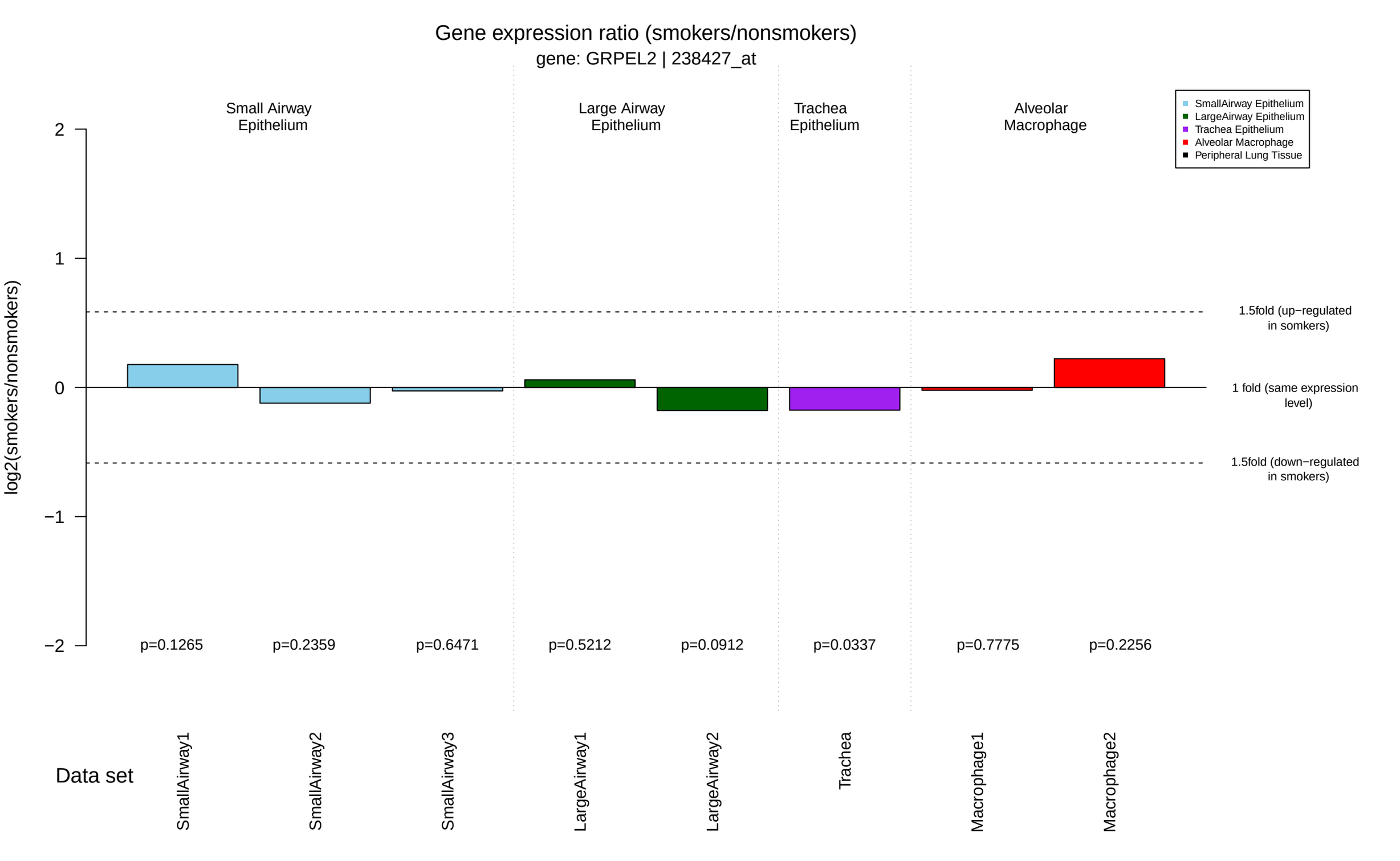
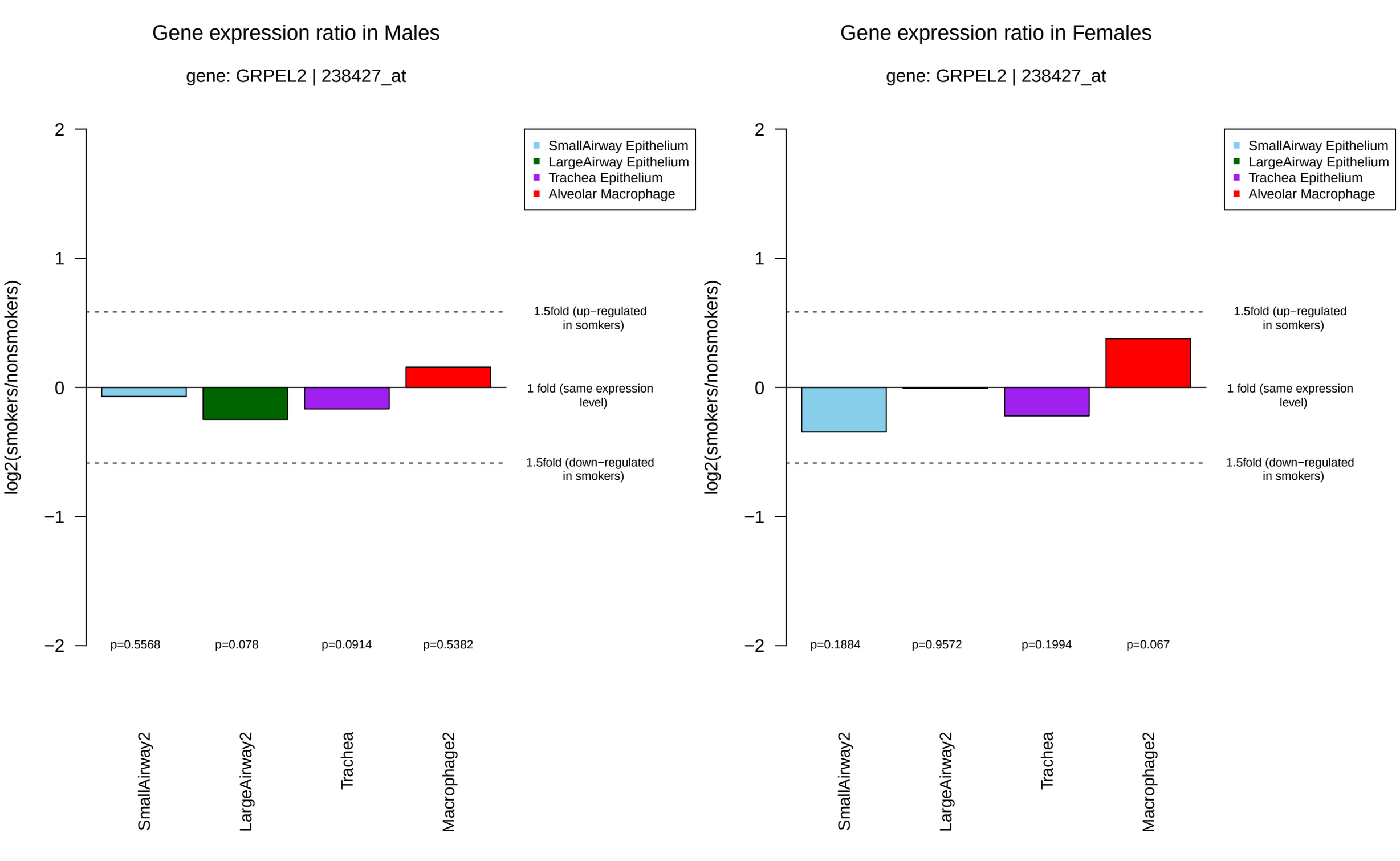
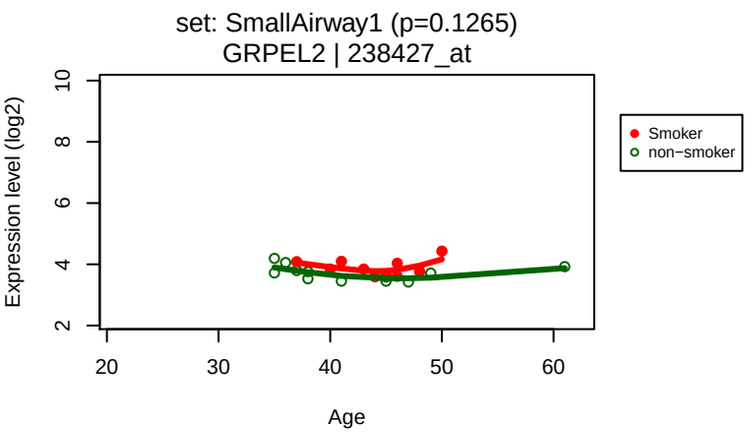
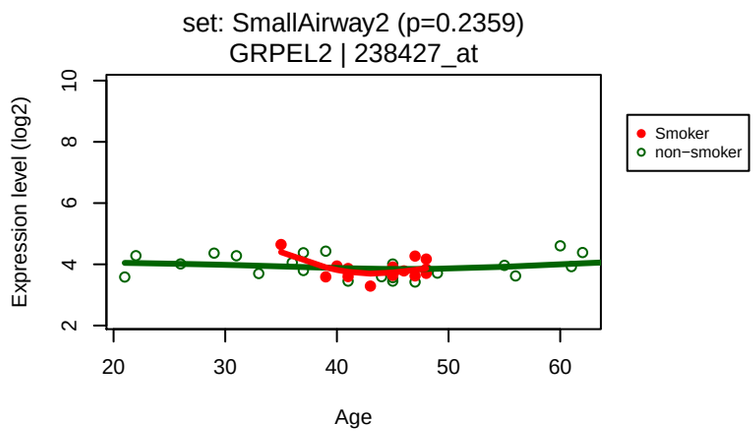
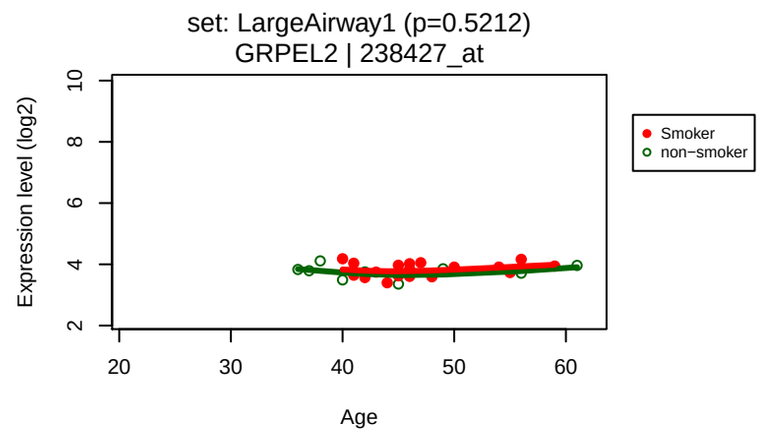

Gene expression of smokers and nonsmokers
gene: GRPEL2 | 238427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1265
p=0.2359
p=0.6471
p=0.5212
p=0.0912
p=0.0337
p=0.7775
p=0.2256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

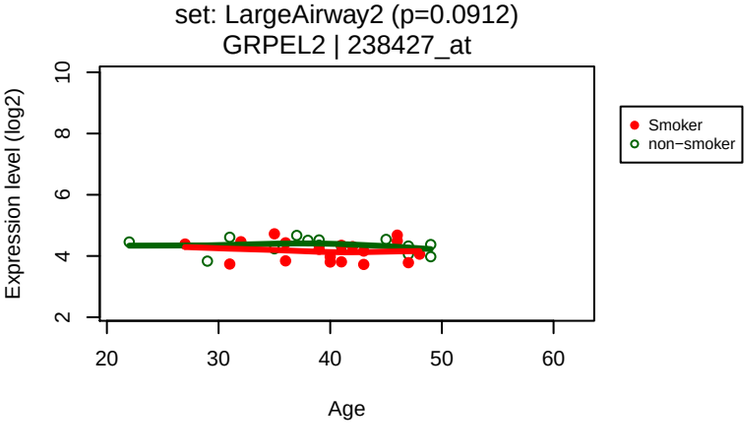

Gene expression ratio (smokers/nonsmokers)
gene: GRPEL2 | 238427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1265
p=0.2359
p=0.6471
p=0.5212
p=0.0912
p=0.0337
p=0.7775
p=0.2256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

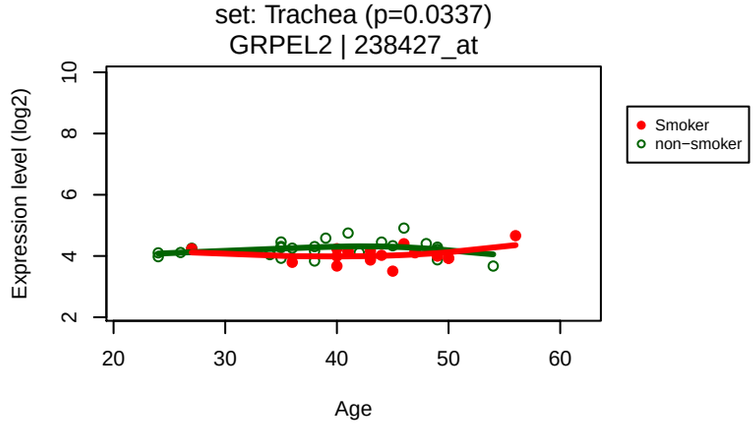

Gene expression ratio in Males
gene: GRPEL2 | 238427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5568
p=0.078
p=0.0914
p=0.5382
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRPEL2 | 238427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1884
p=0.9572
p=0.1994
p=0.067
SmallAirway2
LargeAirway2
Trachea
Macrophage2

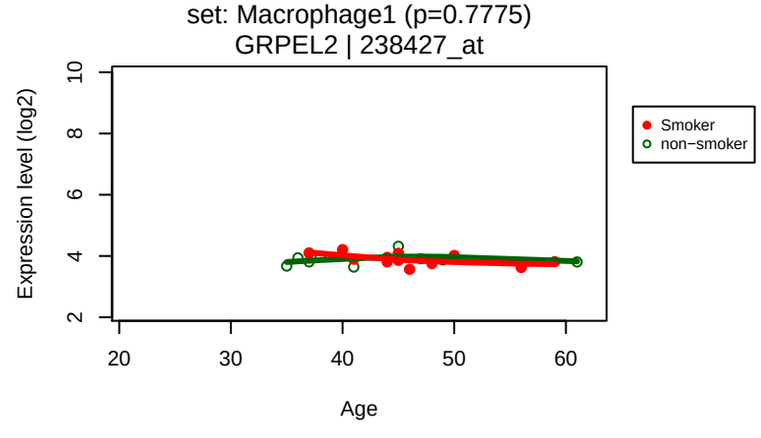

set: SmallAirway1 (p=0.1265)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker

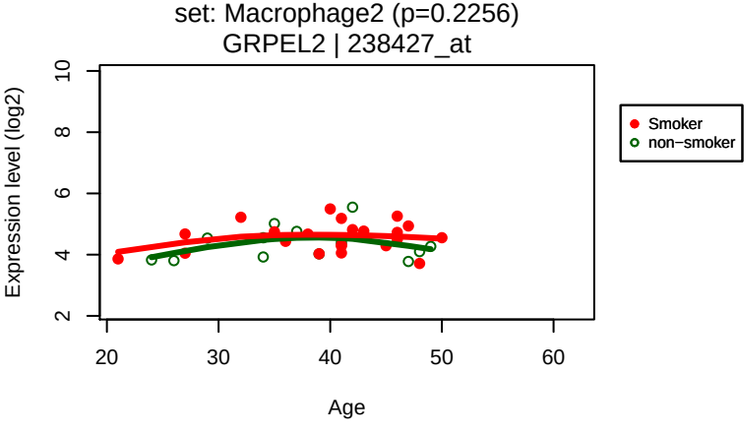

set: SmallAirway2 (p=0.2359)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker

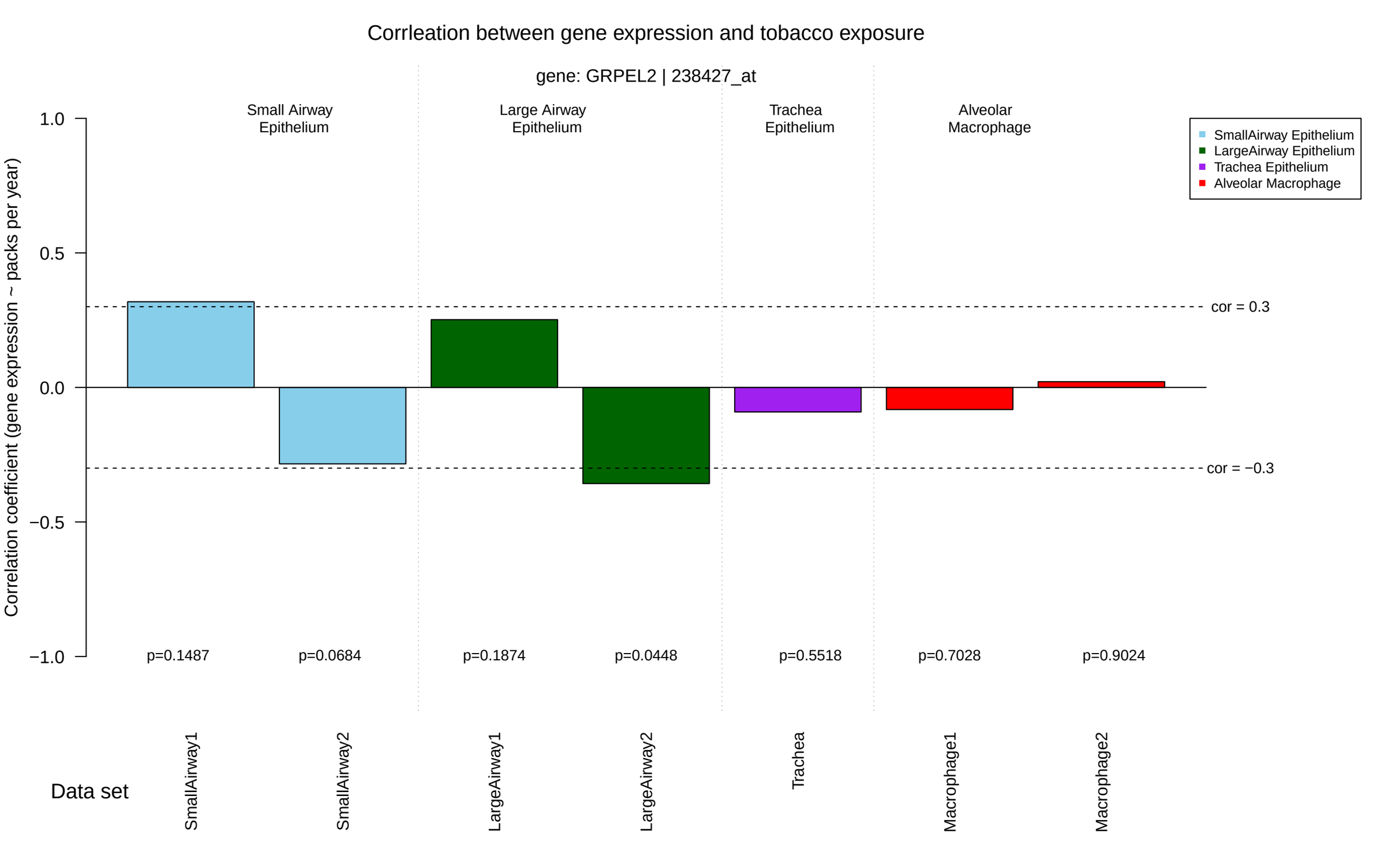

set: LargeAirway1 (p=0.5212)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0912)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0337)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7775)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2256)
GRPEL2 | 238427_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRPEL2 | 238427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1487
p=0.0684
p=0.1874
p=0.0448
p=0.5518
p=0.7028
p=0.9024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2